St. Jude Children's Research Hospital
ALSAC • Danny Thomas, Founder


**Lei Li**
Lead Bioinformatics Research Scientist
St. Jude Children's Research Hospital


## A. PERSONAL DATA


| | |
|---|---|
| Current title | Lead Bioinformatics Research Scientist |
| Affiliation | St. Jude Children's Research Hospital |
| Office address: | 262 Danny Thomas Place, Memphis, TN, 38105, USA |
| Cell phone: | +1 662-694-2218 |
| Email address - Work: | lei.li@stjude.org |
| Email address - Personal: | leilioxford@gmail.com |
| Personal page: | https://leili-uchicago.github.io/ |
| ORCID: | 0000-0002-8828-5199 |
| GitHub: | https://github.com/LeiLi-Uchicago |


## B. RESEARCH INTERESTS

**Single-cell Multi-Omics approaches**, including demultiplexing, clustering, multimodal integration, and immune-repertoire profiling.
**Spatial omics approaches** for studying tissue organization, cellular interactions, and disease-associated microenvironments.
**Cancer biology,** with emphasis on understanding pediatric cancer origins and progression using spatial and multi-omics approaches.
**B-cell immunology and antibody responses,** focusing on vaccination, immunodominance, antigen design, and humoral immunity.
**Evolutionary genetics and viral adaptation,** with applications to influenza, SARS-CoV-2, and respiratory syncytial virus (RSV).

## C. EDUCATION

| | | | |
|---|---|---|---|
| **Ph.D.,** System Engineering | **Xiamen University**<br>Xiamen, China | 2011 - 2015 | 2015 |
| **M.S.**, Control Engineering | **Xi'an Jiaotong University**<br>Xi'an, China | 2009 - 2011 | 2011 |
| **B.S.**, Automation | **Xi'an Jiaotong University**<br>Xi'an, China | 2005 - 2009 | 2009 |

## D. POSTDOCTORAL TRAINING

| | | |
|---|---|---|
| **Postdoctoral Scholar**<br>Bioinformatics and System biology | **The University of Chicago**<br>Advisor: **Dr. Patrick Wilson** | 2019 – 2021 |
| **Postdoctoral Scholar**<br>Bioinformatics and System biology | **Mississippi State University**<br>Advisor: **Dr. Henry Wan** | 2015 – 2018 |

## E. PROFESSIONAL POSITIONS & EMPLOYMENT

| | | |
|---|---|---|
| Lead Bioinformatics Research Scientist | **St. Jude Children's Research Hospital**, Memphis, TN, USA | 2023 – Present |
| Senior Bioinformatics Research Scientist | **St. Jude Children's Research Hospital**, Memphis, TN, USA | 2022 – 2023 |
| Assistant Professor of Bioinformatics Research in Pediatrics | **Weill Cornell Medicine**<br>New York, NY, USA | 2022 – 2022 |
| Senior Bioinformatics Analyst | **Weill Cornell Medicine**<br>New York, NY, USA | 2021 – 2022 |

## F. BIBLIOGRAPHY

# equal contribution * corresponding author

Selected Publications:

1. **Li, Lei#***, Haidong Yi#, Jessica N. Brazelton, Richard Webby, Randall T. Hayden, Gang Wu, and Diego R. Hijano. "*Bridging Genomics and Clinical Medicine: RSVrecon Enhances RSV Surveillance with Automated Genotyping and Clinically-important Mutation Reporting*". ***Influenza and Other Respiratory Viruses*** 20, no. 1 (2026): e70203, DOI: 10.1111/irv.70203
2. Jenna Guthmiller#, Linda Yu-Ling Lan#, **Lei Li#**, Yanbin Fu#, Sean A. Nelson#, Carole Henry, Christopher Stamper et al. "*Long-lasting B cell convergence to distinct broadly reactive epitopes following vaccination with chimeric influenza virus hemagglutinins*". ***Immunity*** (2025) 58, 1–17, DOI:10.1016/j.immuni.2025.02.025
3. **Lei Li#**, Jiayi Sun#, Yanbin Fu#, Siriruk Changrob, Joshua J.C. McGrath, Nai-Ying Zheng, Patrick C. Wilson. "A hybrid demultiplexing strategy for single-cell populations that increases both recovery rate and accuracy". ***Briefings in Bioinformatics*** 25, no. 4 (2024): bbae254, DOI: 10.1093/bib/bbae254
4. **Lei Li**, Yu-Ling Lan, Lei Huang, Congting Ye, Jorge Andrade, and Patrick C Wilson. "Selecting representative samples from complex biological datasets using k-medoids clustering". ***Frontiers in Genetics***, (2022) p. 1787. DOI: 10.3389/fgene.2022.954024
5. **Li, Lei#**, Siriruk Changrob#, Yanbin Fu#, Olivia Stovicek et al. " Librator: a platform for the optimized analysis, design, and expression of mutable influenza viral antigens." ***Briefings in Bioinformatics***, (2022) bbac028, DOI: 10.1093/bib/bbac028

6. **Li, Lei#**, Haley L Dugan#, Christopher Stamper# et al. "Improved integration of single-cell transcriptome and surface protein expression by LinQ-View." ***Cell Reports Methods*** 1, no. 4 (2021): 100056. DOI: 10.1016/j.crmeth.2021.100056
7. Dugan, Haley#, Christopher T Stamper#, **Lei Li#**, Siriruk Changrob et al. "Profiling B cell immunodominance after SARS-CoV-2 infection reveals antibody evolution to non-neutralizing viral targets". ***Immunity*** 54 (6), 1290-1303. e7 (2021). Doi: 10.1016/j.immuni.2021.05.001
8. **Li, Lei**, Deborah Chang, Lei Han, Xiaojian Zhang, Joseph Zaia, Xiu-Feng Wan. "Multi-task learning sparse group lasso: a method for quantifying antigenicity of influenza A (H1N1) virus using mutations and variations in glycosylation of Hemagglutinin." ***BMC bioinformatics*** 21 (2020): 1-22. Doi: 10.1186/s12859-020-3527-5
9. Zhang, Xiaojian#, Hailiang Sun#, Fred L Cunningham#, **Lei Li#**, Katie Hanson-Dorr et al. "Tissue tropisms opt for transmissible reassortants during avian and swine influenza A virus co-infection in swine." ***PLoS pathogens*** 14, no. 12 (2018): e1007417. Doi: 10.1371/journal.ppat.1007417
10. **Li, Lei**, Andrew S Bowman, Thomas J DeLiberto et al. "Genetic evidence supports sporadic and independent introductions of subtype H5 low-pathogenic avian influenza A viruses from wild birds to domestic poultry in North America." ***Journal of virology*** 92, no. 19 (2018). Doi: 10.1128/JVI.00913-18
11. Kosikova, Martina#, **Lei Li#**, Peter Radvak, Zhiping Ye, Xiu-Feng Wan, and Hang Xie. "Imprinting of repeated influenza A/H3 exposures on antibody quantity and antibody quality: implications for seasonal vaccine strain selection and vaccine performance." ***Clinical infectious diseases*** 67, no. 10 (2018): 1523-1532. Doi: 10.1093/cid/ciy327
12. **Li, Lei**, Thomas J. DeLiberto, Mary L. Killian, Mia K. Torchetti, and Xiu-Feng Wan. "Evolutionary pathway for the 2017 emergence of a novel highly pathogenic avian influenza A (H7N9) virus among domestic poultry in Tennessee, United States." ***Virology*** 525 (2018): 32-39. Doi: 10.1016/j.virol.2018.09.003
13. **Li, Lei**, Guoli Ji, Congting Ye, Changlong Shu, Jie Zhang, and Chun Liang. "PlantOrDB: A genome-wide ortholog database for land plants and green algae." ***BMC plant biology*** 15, no. 1 (2015): 1-11. Doi: 10.1186/s12870-015-0531-4

Other Peer-reviewed Research Articles:

14. Fu, Yanbin et al. "In vivo evolution of antibody CR3022 expands cross-neutralization of SARS-CoV-2 variants and informs pan-sarbecovirus immunity." ***Cell Reports*** 45, no. 4 (2026). DOI: 10.1016/j.celrep.2026.117137
15. Sun, Jiayi et al. B cell imprinting in children impairs antibodies to the haemagglutinin stalk. ***Nature*** (2026). DOI: 10.1038/s41586-026-10248-6
16. Ling, Te et al. "GATA1 N-terminus coordinates metabolic reprogramming in erythropoiesis." ***Blood*** (2026). DOI: 10.1182/blood.2025030464
17. Changrob, Siriruk et al. Common cold embecovirus imprinting primes broadly neutralizing antibody responses to SARS-CoV-2 S2. ***Journal of Experimental Medicine*** 222, no. 12 (2025): e20251146. DOI: 10.1084/jem.20251146
18. Seiler, Patrick et al. "Altered germinal center responses in mice vaccinated with highly pathogenic avian influenza A (H5N1) virus." ***Vaccine*** 60 (2025): 127311. DOI: 10.1016/j.vaccine.2025.127311

19. Joshua J.C. McGrath et al. “Mutability and hypermutation antagonize immunoglobulin codon optimality” ***Molecular Cell*** 85, no. 2 (2025): 430-444. DOI: 10.1016/j.molcel.2024.11.033
20. Kim, Youngchang et al. "Epitopes recognition of SARS-CoV-2 nucleocapsid RNA binding domain by human monoclonal antibodies." ***Iscience*** 27, no. 2 (2024). DOI: 10.1016/j.isci.2024.108976
21. Meng Hu et al. “Hemagglutinin destabilizationin H3N2 vaccine reference viruses skews antigenicity and prevents airborne transmission in ferrets.” ***Science Advances*** (2023) DOI: 10.1126/sciadv.adf5182
22. Siriruk Changrob et al. “*Site of vulnerability on SARS-CoV-2 spike induces broadly protective antibody to antigenically distinct Omicron subvariants.”* ***Journal of Clinical Investigation*** (2023) DOI: 10.1172/JCI166844
23. McGrath, Joshua JC et al. "Memory B cell diversity: insights for optimized vaccine design." ***Trends in Immunology*** (2022). DOI: 10.1016/j.it.2022.03.005
24. Jenna J Guthmiller et al. " Broadly neutralizing antibodies target a hemagglutinin anchor epitope." ***Nature*** (2021). DOI: 10.1038/s41586-021-04356-8
25. Siriruk Changrob et al. "Cross-Neutralization of Emerging SARS-CoV-2 Variants of Concern by Antibodies Targeting Distinct Epitopes on Spike". ***Mbio*** 12, no. 6 (2021): e02975-21. DOI: 10.1128/mBio.02975-21
26. Guthmiller, Jenna et al. "First exposure to the pandemic H1N1 virus induced broadly neutralizing antibodies targeting hemagglutinin head epitopes". ***Science Translational Medicine*** (2021) Vol. 13, Issue 596, eabg4535. DOI: 10.1126/scitranslmed.abg4535
27. Rahul Vijay et al. “Hemolysis-associated phosphatidylserine exposure promotes polyclonal plasmablast differentiation”. ***Journal of Experimental Medicine*** 218, no. 6 (2021): e20202359. Doi: 10.1084/jem.20202359
28. Dong-Hun Lee et al. "H7N1 Low Pathogenicity Avian Influenza Viruses in Poultry in the United States During 2018." ***Avian Diseases*** 65, no. 1 (2021): 59-62. Doi: 10.1637/aviandiseases-D-20-00088
29. Fionna A Surette et al. "Extrafollicular CD4 T cell-derived IL-10 functions rapidly and transiently to support anti-Plasmodium humoral immunity." ***PLoS Pathogens*** 17, no. 2 (2021): e1009288. Doi: 10.1371/journal.ppat.1009288
30. Jenna J Guthmiller et al. "An egg-derived sulfated N-Acetyllactosamine glycan is an antigenic decoy of influenza virus vaccines." ***mBio*** (2021). Doi: 10.1101/2021.03.16.435673
31. Jenna J Guthmiller et al. "SARS-CoV-2 infection severity is linked to superior humoral immunity against the spike." ***MBio*** 12, no. 1 (2021). Doi: 10.1128/mBio.02940-20
32. Jenna J Guthmiller et al. "Polyreactive Broadly Neutralizing B Cells Are Selected to Provide Defense against Pandemic Threat Influenza Viruses." ***Immunity*** 53, no. 6 (2020): 1230-1244. Doi: 10.1016/j.immuni.2020.10.005
33. Knight, Matthew et al. "Imprinting, immunodominance, and other impediments to generating broad influenza immunity." ***Immunological Reviews*** 296, no. 1 (2020): 191-204. Doi: 10.1111/imr.12900
34. Meng Hu et al. "HA stabilization promotes replication and transmission of swine H1N1 gamma influenza viruses in ferrets." ***Elife*** 9 (2020): e56236. Doi: 10.7554/eLife.56236
35. Xiaojian Zhang et al. "Tissue tropisms of avian influenza A viruses affect their spillovers from wild birds to pigs." ***Journal of Virology*** 23;94(24):e00847-20. (2020) Doi: 10.1128/JVI.00847-20.

36. Krammer, Florian et al. "Emerging from the shadow of hemagglutinin: neuraminidase is an important target for influenza vaccination." ***Cell host & microbe*** 26, no. 6 (2019): 712-713. Doi: 10.1016/j.chom.2019.11.006
37. Minhui Guan et al. "Aerosol transmission of Gull-Origin Iceland subtype H10N7 influenza A virus in ferrets." ***Journal of virology*** 93, no. 13 (2019). Doi: 10.1128/JVI.00282-19
38. Han, Lei et al. "Graph-guided multi-task sparse learning model: a method for identifying antigenic variants of influenza A (H3N2) virus." ***Bioinformatics*** 35, no. 1 (2019): 77-87. Doi: 10.1093/bioinformatics/bty457
39. Hang Xie et al. "Differential effects of prior influenza exposures on H3N2 cross-reactivity of human postvaccination sera." ***Clinical Infectious Diseases*** 65, no. 2 (2017): 259-267. Doi: 10.1093/cid/cix269
40. Martin, Brigitte et al. "Detection of antigenic variants of subtype H3 swine influenza A viruses from clinical samples." ***Journal of clinical microbiology*** 55, no. 4 (2017): 1037. Doi: 10.1128/JCM.02049-16
41. Huang, Guangzao et al. "Integrating multiple fitting regression and Bayes decision for cancer diagnosis with transcriptomic data from tumor-educated blood platelets." ***Analyst*** 142, no. 19 (2017): 3588-3597. Doi: 10.1039/C7AN00944E
42. Shi, Jieming et al. "mirPRo–a novel standalone program for differential expression and variation analysis of miRNAs." ***Scientific reports*** 5, no. 1 (2015): 1-12. Doi: 10.1038/srep14617
43. Ji, Guoli et al. "PASPA: a web server for mRNA poly (A) site predictions in plants and algae." ***Bioinformatics*** 31, no. 10 (2015): 1671-1673. Doi: 10.1093/bioinformatics/btv004
44. Ye, Congting et al. "detectIR: a novel program for detecting perfect and imperfect inverted repeats using complex numbers and vector calculation." ***PloS one*** 9, no. 11 (2014): e113349. Doi: 10.1371/journal.pone.0113349

In submission and preparation:

45. **Lei Li**, Jiayi Sun, Jenna J. Guthmiller, Yanbin Fu, Christopher T. Stamper, Siriruk Changrob, Nai-Ying Zheng, Min Huang, and Patrick C. Wilson. “Comprehensive and multimodal analyses of B cell and T cell receptor repertoire using VGenes”. Manuscript in preparation.

Reviews and Editorials:

Review research articles for the following journals: ***Genome Biology, Nucleic Acids Research, Briefings in Bioinformatics, Journal of Medical Virology, Giga Science, Journal of Translational Medicine, Frontiers in Immunology, Frontiers in Public Health, Frontiers in Bioengineering and Biotechnology, Frontiers in Molecular Biosciences, Frontiers in Genetics, Frontiers in Veterinary Science, NAR genomics and bioinformatics, BMC bioinformatics*** and ***PloS One***

## G. PRESENTATIONS

1. **Lei Li** et al. “High-resolution spatial omics reveals cellular microenvironment and origins of hepatoblastoma in mouse liver”, presentation at the Pathology Research Symposium 2026, hosted by St. Jude Children’s Research Hospital, Memphis, TN, Apr 10, 2026

2. **Lei Li** et al. “Long-lasting B cell convergence to distinct broadly reactive epitopes following vaccination with chimeric influenza virus hemagglutinins”, presentation at the Knowledge In Data Science 2025 (KIDS25) Meeting, hosted by St. Jude Children’s Research Hospital, Memphis, TN, Oct 28, 2025
3. **Lei Li** et al. “Bridging Genomics and Clinical Medicine: RSVrecon Enhances RSV Surveillance with Automated Genotyping and Clinically-important Mutation Reporting” Poster presentation at IDWeek25 Meeting, Atlanta, GA, Oct, 22, 2025
4. **Lei Li** et al. “Computational Insights into Host-Microbe Interactions”, Invited by Dr. Henry Wan, University of Missouri, Columbia, MO, Mar 07, 2025
5. **Lei Li** et al. “Improving data interpretation for high resolution spatial omics using computational approaches”, Spatial Biology West Coast US 2024, San Diego, CA, December 5, 2024
6. **Lei Li** et al. “Challenges and solutions in high resolution Spatial omics”, Knowledge In Data Science 2024 (KIDS24) Meeting, hosted by St. Jude Children’s Research Hospital, Memphis, TN, Sep 17, 2024
7. **Lei Li** et al. “A hybrid demultiplexing strategy that improves performance and robustness of cell hashing”, Invited by Dr. Yuping Wang, presentation at Tulane University, New Orleans, LA, Online-meeting, Nov 29, 2023
8. **Lei Li** et al. “Using VGenes for an efficient, comprehensive and multimodal analyses of massive B-cell repertoire sequences”, Knowledge In Data Science 2023 (KIDS23) Meeting, hosted by St. Jude Children’s Research Hospital, Memphis, May 15, 2023
9. **Lei Li** et al. “A hybrid demultiplexing strategy for single-cell populations that increases both recovery rate and accuracy”, presentation at the Knowledge In Data Science 2023 (KIDS23) Meeting, hosted by St. Jude Children’s Research Hospital, Memphis, TN, May 15, 2023
10. **Lei Li** et al. “Using Single-cell Multi-omics to Improve the Effectiveness, Efficiency and Economy in Immunology Study”, presentation at the Multi-omics Analysis Interest Group Meeting, co-hosted by Biostatistics department and CAB, St. Jude Children’s Research Hospital, Memphis, TN, Jan 26, 2023
11. **Lei Li** et al. “Understanding B cell dynamics using single-cell sequencing”. Invited by Dr. Yong Cheng, presentation at The Department of Hematology, St. Jude Children’s Research Hospital, Memphis, TN, Sep 15, 2022
12. **Lei Li** et al. “A hybrid demultiplexing strategy for single-cell populations that increases both recovery rate and accuracy”. Invited by Dr. Yifei Xu, presentation at Shandong University, Shandong, China, Online-meeting, Sep 6, 2022
13. **Lei Li** et al. “Comprehensive and multimodal analyses of B cell and T cell receptor repertoire using VGenes”, presentation at Pediatrics Research Day 2022, Weill Cornell Medicine, New York, NY, Jun 2, 2022
14. **Lei Li** et al. “Comprehensive and multimodal analyses of B cell and T cell receptor repertoire using VGenes”, Poster presentation at 2nd Intercampus Immunology Symposium of Cornell University, New York, NY, Apr 19, 2022
15. **Lei Li** et al. “Bioinformatics in vaccine development against viral pathogens”. Invited by Dr. Chun Liang, presentation at Choose Ohio First Seminar, Online-meeting. Feb 10, 2022
16. **Lei Li** et al. “Librator, a platform for optimized sequence editing, design, and expression of influenza virus proteins”. presentation at 2nd Annual CIVICs Network Meeting, New York, NY, Aug 9, 2021

17. **Lei Li** et al. “Librator, a platform for optimized sequence editing, design, and expression of influenza virus proteins”. presentation at Sinai-Emory Multi-Institutional CIVIC (SEM-CIVIC) internal meeting. Online-meeting. March 2021
18. **Lei Li** et al. “Sporadic and independent introductions of low pathogenic H5 avian influenza A viruses from wild birds to domestic poultry in the United States”. presentation at ASV 37th Annual Meeting, College Park, MD, July 2018
19. **Lei Li** et al. “Identification of glycosylation sites and mutations determining antigenic drift events for influenza A viruses using sparse group lasso regression”. Poster presentation at 2016 SFG Annual Meeting, New Orleans, LA, November 2016.
20. **Lei Li** et al. “PlantOrDB: a genome-wide ortholog database for land plants and green algae.” Poster presentation at GLBIO 2014 Meeting, Cincinnati, OH, May 2014